



FNB

APPLICATION PROGRAMMING INTERFACE (API) TRANSACTION HISTORY MESSAGE SPECIFICATION

SOUTH AFRICA





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1.1 Document Scope

The purpose of this document is to provide all details as well as data fields to retrieve Transaction History based on First National Bank (FNB) specific rules and usage guidelines.

1.2 Transaction History Message

This service is for retrieving Transaction History report on the client's account through the API Channel.

Clients would typically request account Transaction History report for a specific date range, account transactions as well as the total balance will form part of the response. The service only allows for one account to be retrieved for Transaction History at a time.

1.3 Referenced Swagger Documents

The Swagger Documents together with the message specification outlining FNB's usage rules should be used as a guideline to retrieve a Transaction History message.

**Reference: For more information about this XSD format please visit www.iso20022.org.*



1.4 Message Specification Explained

The following table explains the usage in the message format.

Column	Definition
Field	Refers to a JSON field.
Name	Refers to the full name of the field.
Multiplicity	This determines if a field is optional or mandatory, and how many times the field can be repeated as per the base ISO message. For example**: [0..1] Shows that the field can appear 0 or 1 time. The field is optional. [0..n] Shows that the field can appear 0 or multiple times. The field is optional. [1..1] Shows that the field is required and must appear once. [1..n] Shows that the field is required and must appear at least once. The field can appear multiple times. In case a lower-level field is required while its higher field is optional, the lower level is required only if the higher-level field is present. Where a field is required in the multiplicity column it will always be required in the ROC column. However, an optional field in the multiplicity column may be overwritten by the rule in the ROC column.
R/O/C	This determines if a field is Required/Optional/Conditional for FNB. R = Required O = Optional C = Conditional. Example: Either the creditor or the ultimate creditor details can be provided. If both are provided the ultimate creditor details will be used. XOR = This field or the next field, but not both must be populated.
FNB Rules	This defines the FNB restrictions on a field. Where length and other format rules differ between ISO default and the FNB length and format rules will be stipulated here.
Additional Information	This provides more information on the usage of the field.



1.5 Transaction History Message Specification

The following table details the structure and format of the fields in the Transaction history request and response.

1.5.1 Query Parameters

Field	Multiplicity	R/O/C	FNB Rules	Additional Information
fromDate	1..1	R	ISODateTime YYYY-MM-DD	Date at which the message was created. Start value for transaction date range. Transaction on, and newer than fromDate will be returned.
toDate	1..1	R	ISODateTime YYYY-MM-DD	End value for transaction date range. Transaction on, and older than fromDate will be returned.
lastItemKey	1..1	C	Text: Maximum = 15	Key used to retrieve the next set of items to be returned. This only required if the lastPageIndicator = "False".



1.5.2 Path Variables

Field	Multiplicity	R/O/C	FNB Rules	Additional Information
accountNumber	1..1	R	Text: Maximum = 23	Client account number that Transaction History is being queried for. Only Allows one account at a time. Must not only contain zeroes or spaces.

1.6 Retrieve Transaction History Report

The below table details the fields in the response message for Transaction History.

Name	Multiplicity	R/O/C	FNB Rules	Additional Information
<u>entry</u>	1..n	R		Specifies an entry in the Transaction History.
entryId	1..1	R	Text: Maximum = 15	The identification field is assigned by the bank and is used to identify each entry in the Transaction History. The field is not unique and can be repeated within the same date range.
bookingDate	1..1	R		Date when an entry is posted to an account on the account servicer's books.
Date	1..1	R	ISODateTime yyyy-mm-dd	Date when an entry is posted to an account on the account servicer's books.



valueDate	1..1	R		Date at which assets become available to the account owner in case of a credit entry or cease to be available to the account owner in case of a debit entry.
Date	1..1	R	ISODatetime yyyy-mm-dd	Date and time at which assets become available to the account owner in case of a credit entry or cease to be available to the account owner in case of a debit entry.
entryDetails		R		Provides further details into the entry.
transactionDetails	1..1	R		Provides information on the underlying transaction(s).
remittanceInfo	1..1	R		Information supplied to enable the matching of an entry with the items that the transfer is intended to settle, such as commercial invoices in an accounts receivable system.
unstructured	1..1	R	Text: Maximum = 30	This is the reference that will appear on the beneficiary's (recipient's) account statement. Payments to LOAN, TRNS, SBSH and GRCP FNB accounts will only process the first 20 characters of this field. Must not only contain spaces. Spaces after text will be trimmed. Where the beneficiary / recipient enforces reference validation on the payment, only the first 20 characters will be validated.
reference	1..1	R		Provides the identification of the underlying transaction.



endToEndId	1..1	R	Text: Maximum = 20	Unique identification, as assigned by the initiating party, to unambiguously identify the transaction. This identification is passed on, unchanged, throughout the entire end-to-end chain.
amount	1..1	R		Value of the transaction and the corresponding currency.
amount	1..1	R	Decimal Number Total Digits: 13 Fraction Digits: 2	Value of the transaction.
currency	1..1	R	Code Text: Maximum = 3	Currency used for the transaction. Currency code as per ISO 4217 three-letter code.
creditDebitIndicator	1..1	R	Code Text: Maximum = 6	Indicates whether the transaction is a credit or a debit transaction. • CREDIT - for all credit transactions. • DEBIT - For all debit transactions.
availability	1..1	R		Identifies the balance available due to that entry.
amount	1..1	R	Decimal Number Total Digits: 13 Fraction Digits: 2	Identifies the value of the available balance in the account.



currency	1..1	R	Code Text: Maximum = 3	Currency used for the transaction. Currency code as per ISO 4217 three-letter code.
creditDebitIndicator	1..1	R	Code Text: Maximum = 6	Indicates whether the availability balance is a credit or a debit balance. Usage: A zero balance is considered to be a credit balance. • CREDIT – for a credit balance. • DEBIT – for a debit balance.
groupHeader	1..1	R		Set of characteristics shared by all individual transactions included in the Transaction History message.
pagination	1..1	R		Provides details on the pagination capabilities of the service.
lastPageIndicator	1..1	R	Boolean True or False	Indicates the last page. • False: This is not the last page of the Transaction History. • True: This is the last page of the Transaction History.
lastItemKey	1..1	R	Text: Maximum = 15	Indicates the last item that has been provided and the next page in the Transaction History can be retrieved from this point. The key will not be used to retrieve the delta transactions but the next batch of transactions within the same range



2. Code Definition Tables

2.1 HTTP Response Status Codes

The below table details the HTTP response status codes sent from the API.

Code	Name	Description
200	Accepted	Request validated successfully, Instruction ID is generated, and Transaction History is returned.
204	No Content	Request has been received and is still being validated and processed in the FNB systems.
400	Bad request	Request sent has failed technical validation.
401	Unauthorised Error	Access token has expired, and a new Access token is required.
403	Forbidden Error	The client is not authorised for the specific service
404	Not found	Resource being called cannot be found at the given endpoint.
429	Too Many Requests Error	Too many requests have been sent.
500	Internal Server Error	The server encountered an unexpected condition that prevented it from fulfilling the request.
503	Service Unavailable	API service is unavailable.



2.2 Permitted Character Set

The below table details the character sets that are applicable in the request.

Character	Description
"A" – "Z"	29 capital characters of the Latin alphabet (Uppercase)
"a" – "z"	29 capital characters of the Latin alphabet (Lowercase)
"0" – "9"	10 numeric characters
"."	Period
"_"	Hyphen
"*"	Asterisk
","	Comma
"("	Left parenthesis
")"	Right parenthesis
"%"	Percentage
"+"	Plus
"\$"	Dollar
","	Semi-colon
"="	Equal
"@"	At
"?"	Question mark
":"	Colon
" "	Space
"!"	Exclamation mark
"	Quotes
"#"	Hash
"&"	Ampersand
'	Quote
"/"	Slash
"<"	Less than
">"	Greater than
"["	Square bracket (left)



"\"	Back slash
"]"	Square bracket (right)
"^"	Circumflex
" _"	Underscore